AI-Powered Project Triage & Engineering Impact Assessment

Role: End-to-end system design and implementation | Deployment: Production

Executive summary

I built a production AI-assisted triage workflow to accelerate the first-pass assessment of incoming engineering initiatives. The system combines structured intake, deterministic screening, retrieval of product and engineering context, and LLM-based analysis. It returns a reviewable impact assessment with suggested teams, effort ranges, evidence, assumptions, and confidence signals.

Scale and design at a glance

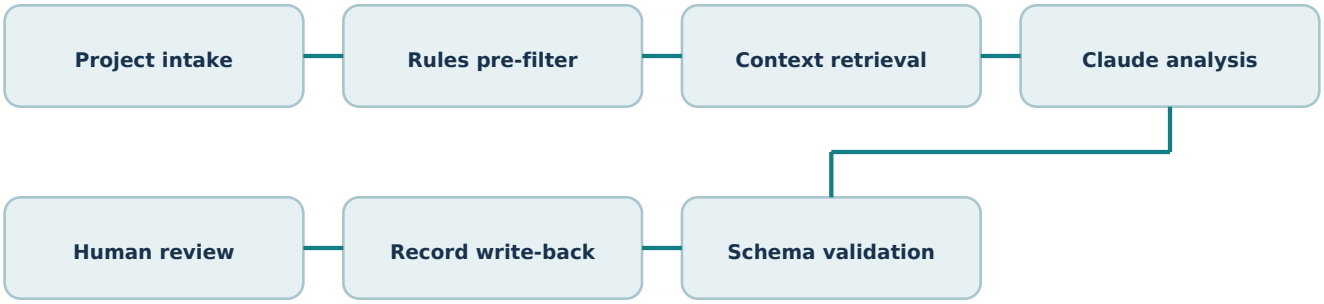
800–900	3	5	6
Projects in the broader planning cycle*	Impact outcomes: yes / no / maybe	Distinct confidence dimensions	Projects in a documented early pilot

*Planning-cycle volume describes the surrounding operating environment, not a measured count of records processed automatically. The six-project pilot preceded the confirmed production deployment.

The problem

A high-volume planning process required engineering teams to assess project requirements, infer affected systems and teams, locate relevant historical work, and estimate rough effort. Manually assembling this context across intake records, documents, and issue trackers created repeated work and inconsistent recommendations.

Solution architecture



Context retrieval: relevant product requirements and historical engineering tickets were fetched asynchronously through knowledge and issue-tracking integrations. The output was validated against a typed schema before write-back.

Key technical design choices

Deterministic pre-filter: Screens obvious cases before invoking retrieval and model inference.

Bounded asynchronous retrieval: Reduces serial waiting while enforcing timeouts and concurrency limits.

Structured output contract: Validates impact, team recommendations, sizing, supporting evidence, and uncertainties.

Evidence-backed confidence: Separates facts from assumptions and provides five confidence dimensions rather than a single opaque score.

Human-in-the-loop review: Keeps final impact and sizing decisions with engineering owners.

Representative triage output

Output field	Illustrative result
Impact	Maybe — engineering review required
Potential teams	Platform services; customer experience
Sizing	Range-based initial estimate; owner confirmation required
Evidence	Requirement excerpt + comparable historical tickets
Uncertainty	API dependency and scope boundary require clarification

The example above uses fictional project content; it illustrates output structure, not a real assessment.

Production outcome

The system was fully working and used in production. It established a repeatable, evidence-linked first-pass assessment workflow and supported consistent human review of engineering impact and effort. I owned the technical design and implementation, including integration orchestration, validation, and operational controls.

Measurement integrity: Planning volume, pilot size, output categories, and confidence dimensions are supported by the project materials. No unsupported time-savings, accuracy, or throughput improvement is claimed.